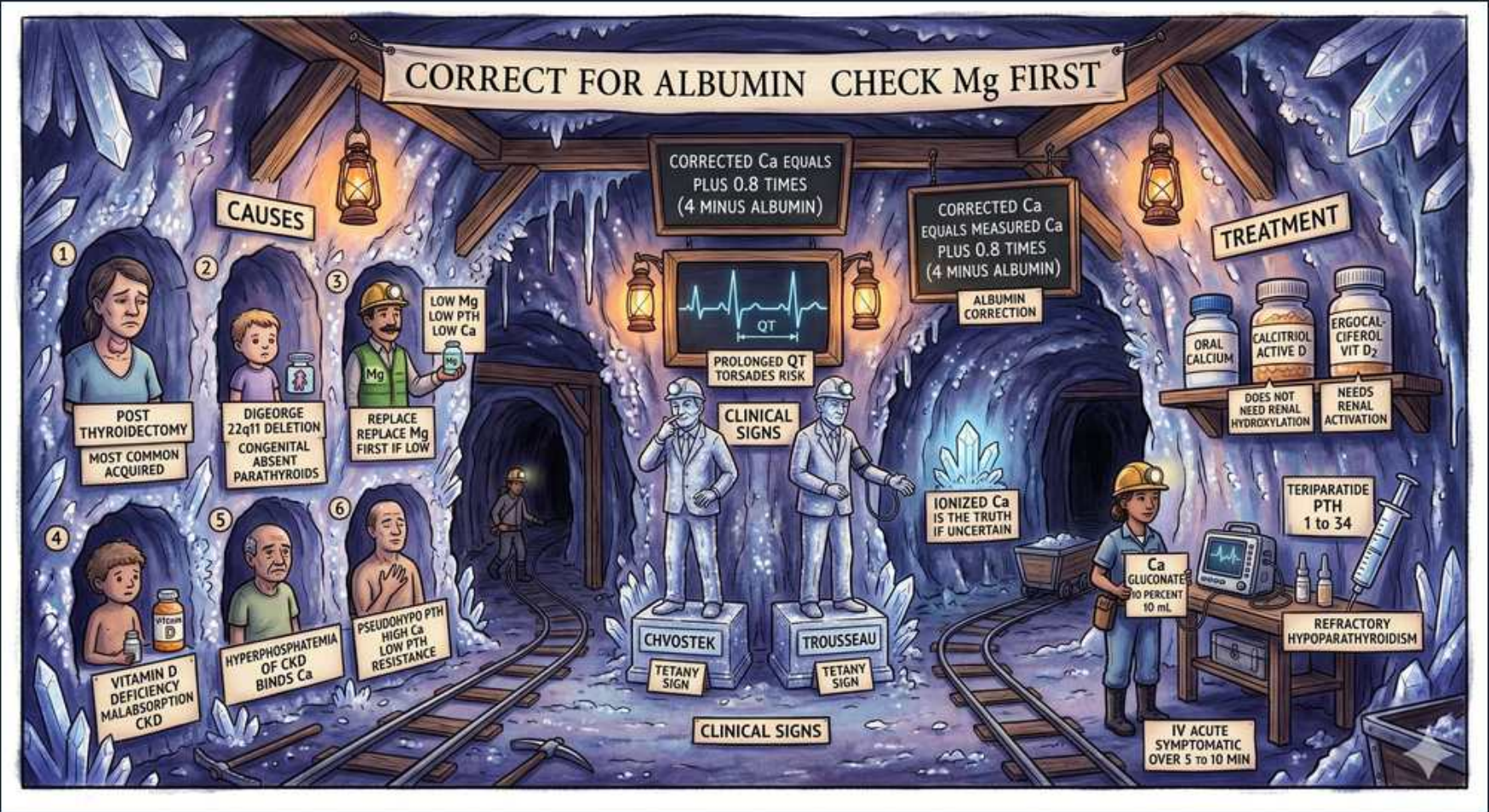


Calcium (Renal) — Hypocalcemia



1. Correct for albumin

WHAT YOU SEE

Banner: CORRECT FOR ALBUMIN, CHECK Mg FIRST

WHAT IT ENCODES

Two reflexes before treating any low calcium: (1) correct total Ca for albumin or get an IONIZED calcium (~50% of calcium is albumin-bound), and (2) check MAGNESIUM. Hypoalbuminemia (cirrhosis, nephrosis, critical illness) is the most common cause of a 'low' total calcium with a normal ionized fraction — no treatment needed.

BOARD TRAP

WRONG: giving calcium for a low TOTAL calcium in a hypoalbuminemic patient. Their ionized (active) calcium is normal — this is pseudohypocalcemia. Correct for albumin or measure ionized Ca first.

2. Albumin correction formula

WHAT YOU SEE

Corrected Ca = measured + 0.8 x (4 - albumin)

WHAT IT ENCODES

Corrected Ca (mg/dL) = measured Ca + 0.8 × (4.0 – serum albumin in g/dL). Each 1 g/dL drop in albumin below 4 lowers measured calcium by ~0.8 mg/dL without changing the physiologically active ionized fraction. When in doubt (acid-base shifts, critical illness), an IONIZED calcium beats the formula.

BOARD TRAP

WRONG: trusting the correction formula in unstable critically-ill patients or significant acid-base derangement. The formula is unreliable there — order a direct ionized calcium instead.

3. Post-thyroidectomy

WHAT YOU SEE

Cause panel #1: a surgically-bandaged neck figure labelled POST THYROIDECTOMY / MOST COMMON ACQUIRED — the cut neck means the parathyroids were accidentally removed (low PTH).

WHAT IT ENCODES

Post-surgical hypoparathyroidism (thyroidectomy, parathyroidectomy, radical neck) is the #1 ACQUIRED cause: inadvertent removal/devascularization of parathyroids → LOW PTH, LOW Ca, HIGH phosphate. Can be transient or permanent. Monitor calcium closely for 24–72 h post-op; treat with calcium + calcitriol.

BOARD TRAP

WRONG: confusing post-op hypoparathyroidism with HUNGRY BONE syndrome. Hypoparathyroidism gives HIGH phosphate (low PTH); hungry bone gives LOW phosphate (avid bone mineral uptake) — the phosphate level distinguishes them.

4. DiGeorge 22q11

WHAT YOU SEE

Cause panel #2: a child figure labelled DIGEORGE 22q11 DELETION / CONGENITAL ABSENT PARATHYROIDS — the missing-gland baby encodes congenital hypoparathyroidism.

WHAT IT ENCODES

DiGeorge syndrome (22q11.2 deletion) → failure of 3rd/4th pharyngeal pouch development → congenitally ABSENT parathyroids and thymus: hypocalcemia (low PTH) with tetany/seizures, T-cell immunodeficiency, conotruncal cardiac defects, and cleft palate. CATCH-22 mnemonic. Presents in infancy with hypocalcemic seizures.

BOARD TRAP

WRONG: missing the immunodeficiency/cardiac context. A neonate with hypocalcemic seizures, recurrent infections, and a conotruncal heart defect (e.g., tetralogy/truncus) is DiGeorge, not isolated hypoparathyroidism.

5. Check Mg first

WHAT YOU SEE

Cause panel #3: a low-Mg figure labelled REPLACE Mg FIRST — low magnesium shuts down PTH secretion/action, so the Mg bottle must be given before calcium will rise.

WHAT IT ENCODES

Hypomagnesemia (alcoholism, PPIs, diarrhea, diuretics) causes FUNCTIONAL hypoparathyroidism: low Mg both impairs PTH SECRETION and induces PTH RESISTANCE → hypocalcemia. The calcium will NOT correct until magnesium is repleted. Always check and replace Mg in any refractory hypocalcemia.

BOARD TRAP

WRONG: repeatedly giving calcium for hypocalcemia that won't budge while ignoring magnesium. Calcium alone fails when Mg is low — replete magnesium first and the calcium follows.

6. Vitamin D deficiency / CKD

WHAT YOU SEE

Cause panel #4: a figure with a vitamin-D bottle labelled VITAMIN D DEFICIENCY / MALABSORPTION / CKD — low vitamin D drops gut calcium, driving a compensatory high PTH.

WHAT IT ENCODES

Vitamin D deficiency (poor intake/sun, malabsorption, liver disease) and CKD lower calcium with a COMPENSATORY HIGH PTH (secondary hyperparathyroidism). In CKD, failing kidneys cannot make 1,25-D and retain phosphate → low Ca, high phosphate, high PTH; treat with calcitriol/analogs and phosphate binders.

BOARD TRAP

WRONG: interpreting the elevated PTH here as a parathyroid problem. This is APPROPRIATE secondary hyperPTH responding to low calcium — fix the vitamin D/phosphate, don't operate on the parathyroids.

7. Hyperphosphatemia

WHAT YOU SEE

Cause panel #5: a figure labelled HYPERPHOSPHATEMIA / BINDS Ca / CKD — excess phosphate clamping onto calcium (Ca-PO₄ complexes) pulls down ionized calcium.

WHAT IT ENCODES

Hyperphosphatemia (CKD, tumor lysis syndrome, rhabdomyolysis, phosphate enemas) binds calcium into calcium-phosphate complexes, acutely lowering ionized calcium. In tumor lysis, the high phosphate × calcium product risks metastatic calcification — so you do NOT aggressively give calcium unless symptomatic.

BOARD TRAP

WRONG: aggressively repleting calcium in tumor lysis/hyperphosphatemic hypocalcemia. Adding calcium to a high phosphate raises the Ca×PO₄ product and causes precipitation/calcification — treat the phosphate, give calcium only for symptomatic/severe hypocalcemia.

8. Pseudohypoparathyroidism

WHAT YOU SEE

Cause panel #6: a round-faced short figure labelled PSEUDOHYPO PTH / HIGH PTH / RESISTANCE — the Albright body habitus with high PTH encodes end-organ PTH resistance.

WHAT IT ENCODES

Pseudohypoparathyroidism (type 1a, GNAS/Gs-alpha defect, Albright hereditary osteodystrophy) = end-organ RESISTANCE to PTH → low Ca, high phosphate, but HIGH PTH. Phenotype: short stature, round face, shortened 4th/5th metacarpals, obesity. The kidney fails to respond to PTH despite high levels.

BOARD TRAP

WRONG: expecting LOW PTH because the picture mimics hypoparathyroidism. In pseudohypoparathyroidism PTH is HIGH (resistance) — the elevated PTH with low Ca/high phosphate plus the Albright phenotype is the giveaway.

9. Prolonged QT / torsades

WHAT YOU SEE

ECG monitor, PROLONGED QT, torsades

WHAT IT ENCODES

Hypocalcemia prolongs the ST segment and QT interval (phase 2 plateau lengthening) → risk of torsades de pointes and ventricular arrhythmia. Severe cases also cause laryngospasm, bronchospasm, and seizures. Symptomatic/prolonged-QT hypocalcemia is an emergency requiring IV calcium.

BOARD TRAP

WRONG: assigning a SHORT QT to hypocalcemia. SHORT QT = HYPER-calcemia; LONG QT = HYPO-calcemia. Reversing them is a classic ECG trap.

10. Chvostek sign

WHAT YOU SEE

Miner face tap, CHVOSTEK, tetany

WHAT IT ENCODES

Chvostek sign: tapping over the facial nerve (anterior to the ear) triggers ipsilateral facial muscle twitching — a sign of NEUROMUSCULAR IRRITABILITY from hypocalcemia. Reflects latent tetany. Less specific than Trousseau (can be positive in ~10% of normal people).

BOARD TRAP

WRONG: relying on a positive Chvostek alone to diagnose hypocalcemia. It occurs in a minority of normocalcemic people — confirm with calcium/ionized levels; Trousseau is the more specific bedside sign.

11. Trousseau sign

WHAT YOU SEE

BP cuff hand spasm, TROUSSEAU, tetany

WHAT IT ENCODES

Trousseau sign: inflating a BP cuff above systolic for ~3 minutes induces CARPAL SPASM (flexed wrist/MCPs, extended fingers, adducted thumb) — more SPECIFIC for hypocalcemia than Chvostek. Reflects increased neuromuscular excitability from low ionized calcium.

BOARD TRAP

WRONG: treating a normal total calcium as excluding tetany. Alkalosis (hyperventilation) lowers ionized calcium with normal total Ca and can produce a positive Trousseau — check ionized calcium and pH.

12. Ionized if uncertain

WHAT YOU SEE

Blue crystal: IONIZED Ca is the truth if uncertain

WHAT IT ENCODES

Ionized calcium is the physiologically active fraction and the GOLD STANDARD when albumin is abnormal, acid-base status is shifting, in critical illness, massive transfusion (citrate), or whenever the corrected-Ca formula is unreliable. It directly reflects what the patient's tissues experience.

BOARD TRAP

WRONG: defaulting to total calcium in the ICU/transfused/alkalotic patient. The total (and even corrected) value can mislead — order an ionized calcium to know the truth.

13. Oral Ca + calcitriol

WHAT YOU SEE

Oral calcium + calcitriol (active vit D) bottles

WHAT IT ENCODES

Chronic/non-emergent hypocalcemia: oral calcium (carbonate or citrate) PLUS active vitamin D (CALCITRIOL, 1,25-D) — calcitriol is needed because hypoparathyroid/CKD patients cannot 1-alpha-hydroxylate plain vitamin D. Goal is low-normal calcium to avoid hypercalciuria/stones; monitor urine calcium.

BOARD TRAP

WRONG: giving plain cholecalciferol/ergocalciferol alone in hypoparathyroidism or CKD. Without PTH (or renal 1-alpha-hydroxylase), inactive vitamin D can't be converted — use ACTIVE calcitriol.

14. IV Ca gluconate if severe

WHAT YOU SEE

Ca gluconate vial, IV acute symptomatic, over 5-10 min

WHAT IT ENCODES

Severe/symptomatic hypocalcemia (tetany, seizures, laryngospasm, prolonged QT/arrhythmia) → IV CALCIUM GLUCONATE (1–2 g over 10–20 min, then infusion). Gluconate is preferred peripherally; calcium CHLORIDE has 3× the elemental calcium but is more sclerosing and needs central access.

BOARD TRAP

WRONG: pushing IV calcium rapidly, or giving calcium chloride peripherally. Fast administration causes arrhythmia/bradycardia (especially on digoxin), and CaCl₂ extravasation causes tissue necrosis — infuse slowly via the appropriate line.